

Integrated Study and Research
Repository with Flashcard-Based
Learning

Richard Egenti 001239105

Nageena Frost

Initial Contextual Report

COMP1682 Final Year Project

BSc H COMPUTING (EXT)

Due Date 20/04/2026

Word Count: 15000

1. Table of Contents
2. Introduction
 - 2.1 Background

University students are increasingly challenged by dual demands of managing large volumes of academic material and retaining key information for long term use. As higher education shifts to digital learning environments, the volumes of notes have grown substantially through articles, blogs, books and more being available online, this has often led to disorganisation and information

overload for higher education students. Sillence et al. (2022) found that while students have found various ways to manage their university-based data there are still many who experience difficulties with information overload. Similarly, Arden et al. (2024) highlights how students adapt to digital note-taking practices in lectures but also face different challenges trying to effectively integrate them into the learning routines. These findings suggest that there is a need for more effective systems that support both resource management and effective study practices.

Also, research in educational psychology consistently demonstrates that practices based on spaced repetition are among the most successful practices for improving long term retention. Varkey et al. (2024) showed that a mobile flashcard application consisting of spaced repetition significantly enhanced dental students' long-term knowledge compared to more traditional methods. Similarly, Lu et al. (2021) reported that the use of the flashcard application Anki was effective for better learning efficiency and stronger in-depth medical knowledge.

This project attempts to help higher education students overcome two typical challenges - knowledge retention and information overload. It will accomplish this by creating a web-based program that combines a research repository and a flashcard learning system. Academic materials saved in the database may be accessed directly via the platform, and each source will be connected to automatically created flashcards that emphasise key points or subjects. When students examine

these flashcards, selecting a certain point directs them to the matching section of the original source. This study aims to improve the efficiency and engagement of students in organising, accessing, and retaining academic material by integrating systematic resource management with active recall.

2.2 Aims and Objectives

The aim of this project is to develop a web-based application that combines a research library with flashcards to assist students in organising academic resources, extracting key concepts, and practicing active recall. Therefore, improving information retention, reducing overload, and encouraging better study habits.

3.1.1 Research into existing research repository tools and flashcard systems, evaluating their functionality and limitations through academic sources. [5.0]

3.1.2 Write a literature review on the importance of information management systems and active recall methods in higher education, with clear analysis of assumptions and limitations. [12.0]

3.1.3 Research functional and non-functional requirements for an integrated repository and flashcard system, based on insights from the literature review and user expectations. [3.0]

3.1.4 Verify requirements by designing and distributing a questionnaire or short survey to students on their study habits and tool preferences. [5.0]

3.1.5 Create a rich picture diagram showing students, their study workflows, and how repository and flashcard features interact. [3.0]

3.1.6 Create use-case diagrams illustrating system functionalities such as uploading resources, generating flashcards, and reviewing content. [3.0]

3.1.7 Create a conceptual ERD to outline the entities and their relationships. [3.0]

3.1.8 Create a physical ERD diagram specifying attributes, keys, and database schema structure. [6.0]

3.1.9 Design possible user interfaces using tools such as Figma or Axure RP. [7.0]

3.1.10 Implement the database schema by creating tables and relationships in MySQL or MongoDB. [10.0]

3.1.11 Develop the main user interface using a web framework using PHP and Html. [13.0]

3.1.12 Implement the resource upload and organisation features, allowing users to

add, tag, and search academic sources. [8.0]

3.1.13 Implement flashcard generation linked to stored resources, displaying key concepts or topics. [13.0]

3.1.14 Create functionality where clicking a flashcard topic redirects users to the relevant section in the source. [8.0]

3.1.15 Implement user registration and login functionality with secure authentication. [8.0]

3.1.16 Add an admin panel to manage users, resources, and system data. [8.0]

3.1.17 Usability testing to observe how easily students navigate the repository and flashcard features. [5.0]

3.1.18 Black box testing to ensure system functions work as expected for users unfamiliar with the code. [5.0]

3.1.19 White box testing to verify system logic and database interactions for flashcard generation and resource retrieval. [5.0]

3.1.20 Evaluate whether the system meets the initial functional and non-functional requirements. [5.0]

3.1.21 Assess the system's usefulness in reducing information overload and improving knowledge retention through small-scale user feedback. [5.0]

3.1.22 Discuss limitations, challenges, and potential improvements for future development. [5.0]

2.3 Approach

This project takes a functional and progressing approach combining academic research with practical systems designing and prototyping. This project's key objective is to create and develop a functional prototype that targets a real-world problem experienced by university students, which is the difficulty of structuring academic information and retaining key information. As a result, the strategy emphasises slow development, constant refinement and the alignment of research findings and system features through strict adherence to a formal branded software development process.

The project begins with a research-informed phase, which includes a methodical evaluation of academic literature digital resource management, flashcard based learning and spaced repetitio. This stage identifies frequent student challenges, such as information overload, inefficient material organisation and insufficient long-term retainment of key information. The findings of the literature review immediately affect the goal and direction of the project, ensuring that the proposed solution is based on current research rather than entirely subjective assumptions. The research-based methodology helps to justify the inclusion of essential elements like the study repository, flashcard production and the clear combination of learning and revision materials.

Following the literature review, the project will use an iterative and continuous development method, rather than attempting to design and build the full system in one step. Initial iterations concentrate on establishing key features, such as user authentication, fundamental repository administration and manual flashcard production. Subsequent production expands on this foundation by offering more complex capabilities like greater organising, filtering and the ability to link flashcards to specific academic sources. This technique allows for constant development while minimising the risk associated with excessively complicated early design decisions.

Prototyping is a key component of the selected agile methodology. Low and mid fidelity designs and prototypes are used to test layout, navigation and user interface, allowing the design concepts to be tested prior to final implementation. As the project advances through revisions, these prototypes transform into working versions of the system. This incremental prototyping approach encourages continued testing and improvement, allowing usability concerns to be detected and resolved early in the development cycle. It also guarantees that design decisions are adaptable and sensible to new requirements and issues that arise throughout the project.

The implementation phase focuses on creating a web-based database driven prototype that makes use of relevant data storage and system logic technologies. The usefulness of technology for handling structured academic data is impacted by existing data, allowing the project to focus on system designs and functionality rather than spending too much time learning unfamiliar tools.

Evaluation and reflection are used throughout the project's lifespan. The system is constantly evaluated against the established functional and non-functional requirements to see whether it accomplishes its intended goals. Emphasis is placed on usability, resource organisational efficacy, and the systems capacity to facilitate learning and revision. Reflection on both technical and design decision informs the future development and gives critical assessment of the final prototype's strengths and limitations.

Overall, this method promotes the creation of a realistic, research-based prototype while allowing for adequate flexibility to respond to problems and altering understanding during the project.

2.4 Justification of methodology or framework

The Agile methodology is adopted for this project because it supports a user-friendly approach and an extremely flexible approach to system development, which is essential for building the learning platform that integrates a research repository and flashcards. The process enables the project to move through organised but adaptive stages of design, implementation, testing,

and evaluation, guaranteeing that each system component is constantly modified in response to user feedback.

During the design stage, Agile facilitates the construction of user-centered interface prototypes, such as the repository dashboard, flashcard viewer, and source-linking functions, which are jointly built to meet student demands. The implementation and development stages are divided into small,

manageable sprints, allowing for incremental improvements and rapid adjustments depending on testing findings. Although this project is being created independently, Scrum concepts will be

modified for solo usage by considering each sprint as a concentrated development period with well-defined goals, such as design, implementation, and testing. Progress will be tracked via self-review at the conclusion of each sprint, allowing for reflection, refinement, and job prioritisation in the following phase.

Frequent testing ensures that the interface stays intuitive and that data retrieval and flashcard creation work properly. Finally, the evaluation step compares the project outputs to the original objectives, focusing on usability, functionality, and improved knowledge retention.

According to Santos et al. (2024), Agile practices such as Scrum use adaptability, collaboration and SMART project outcomes within academic environments. Similarly, Friess (2023) found that groups using Scrum achieved higher efficiency through better communication and better performance

scores compared to traditional approaches. These findings support the use of agile for this project, as it allows for continued refinement to ensure that the final product meets students' needs for information organisation and long-term knowledge retention.

3. Literature review

3.1 Approach to literature searching

The literature review was completed using a structured search approach to identify relevant academic resource management, and flashcard-based learning. The main aim of the literature search was to establish a strong theoretical foundation for the project and guarantee that the proposed system is formed by existing research rather than assumptions.

Google scholar was the primary source for identifying literature, due to its extensive coverage of peer reviewed journal articles, conference papers, and academic publications across computing, education and information systems. In addition to Google Scholar, academic publisher platforms and digital libraries commonly used within computing and educational technology research were consulted, including sources that host peer-reviewed journals and conference proceedings. These platforms were selected because they provide access to credible, research-based material appropriate at the undergraduate level.

Search terms were chosen to represent the project core themes and adjusted iteratively as the scope of the study became clearer. Keywords and phrases included combinations of terms such as digital study tools, academic resource organisation, student information overload, flashcard learning, retrieval practice, spaced repetition and technology-supported learning. Boolean operators were used where appropriate to restrict results and improve relevance. This iterative refinement ensured that the search process focussed on material directly relevant to the projects aims and objectives.

To maintain relevance and academic quality, inclusion and exclusion criteria were applied during the selection process. The priority was peer reviewed journal articles and conference papers published within the last five years to represent the current research and technological practices. However, a small number of older studies were cited because they provided foundational theoretical concepts relevant to learning and memory. Studies prioritised on higher education environments, whereas non-academic sources, opinion pieces and articles lacking empirical or theoretical foundation were removed.

Once identified the relevant material was evaluated and organised thematically and chronologically. Reference management software was used to store sources, manage citations and generate a bibliography in the Harvard referencing style format. This

approach supported consistency, reduced referencing errors and enabled efficient management of a growing body of literature.

Overall, this methodical approach to literature searching guaranteed that the evaluation was thorough, relevant and academically rigorous, providing a strong foundation for identifying the problem and informing subsequent requirements and design of the proposed system.

3.2 Identifying the problem

The literature identifies several interrelated issues that university students encounter as learning where more learning takes place in a digital environment. According to Sillence et al. (2022), one of the most significant challenges encountered is students' difficulty in organising and managing large amounts of academic information. As digital technologies have become significantly more important in higher education, students are required to engage with large volumes of lecture material, academic papers, online resources and personal notes. While these technologies offer improved access to knowledge, research indicates that they have also contributed to information overload and fragmented study habits.

Sillence et al. (2022) shows how students accumulate academic data across multiple platforms frequently without a clear organisational strategy. Their findings indicate that students frequently have trouble identifying crucial material when rewriting, resulting in inefficiencies and frustration. The study emphasises that although students make considerable use of digital technologies, these tools do not always support effective long-term retrieval of information. This is especially problematic during assessment periods, where the ability to acquire and synthesise key information is crucial.

In addition to organisational challenges, the literature indicates that digital learning environments often rely on a fragmented ecosystem of technologies. Bond et al. (2018), in their study of digital transformation in higher education, highlights that learning materials are usually implemented in isolation rather than part of an integrated system. Students may use one platform for storing key information, another for storing resources and a separate tool for revision or testing. This framework can impair learning continuity and increase cognitive load, and students must constantly switch contexts and mentally link knowledge across systems.

The separation of information storage and active learning is a recurring concern in the literature. While repositories and learning management systems facilitate information distribution and storage, they do not promote deep learning or retention. Bond et al. (2018), argues that many digital learning tools emphasise access and delivery above meaningful engagement, which can lead to surface level learning habits. This discrepancy implies that effective learning requires not only access to information but also structured opportunities for active engagement with that information.

Research into learning strategies provides further insight into this problem. In their systematic evaluation of interleaving as a concept learning strategy, Firth et al. (2021) emphasises the need for spreading and diversifying learning sessions to promote deeper knowledge retention and comprehension. Their findings indicate that organised learning strategies such as interleaving and spacing, can significantly increase learning outcomes when compared to massed or repetitive study. However, the review also shows that such strategies are not consistently embedded within everyday study practices or supported by frequently used digital tools.

Trumble et al. (2023), emphasises the advantages of retrieval practice and distributed learning systems. Their thorough research shows strategies that requires learners to actively recall information for better long-term retention and knowledge transfer. Despite this strong evidence base, Trumble et al. (2023) observes that retrieval-based strategies are often implemented from students overall learning processes.

A further issue identified in the literature is the absence of an explicit relationship between revision material and its academic source. Firth et al. (2021) and Trumble et al. (2023) argue that learning procedures are more effective when students can relate recalled material to its original context. However, most digital study tools lack systematic assistance for integrating revision activities to specific lecture notes, reading or research papers. This increases cognitive demands on students and may discourage long term engagement with evidence-based learning practices.

When taken together, these studies demonstrate a clear problem space. Students are facing increasing information overload by relying on fragmented digital tools and frequently lack structured assistance for incorporating good learning habits into their study routines (Sillence et al., 2022; Bond et al., 2018). Although research strongly supports retrieval practice, spacing and interleaving, existing systems rarely integrate these approaches with effective resource management or contextual linking (Firth et al., 2021; Trumble et al., 2023).

This gap directly influences the scope of the planned project. According to the research, there is a need for an integrated system that supports both academic resource organisation and the implementation of evidence-based learning practices in one educational environment. By addressing issues of fragmentation, information overload, and weak contextual connections between learning materials and revision activities, the project hopes to directly address the identified challenges while also laying the groundwork for the requirements and design decisions discussed in subsequent chapters.

3.3 Repository systems in higher education: Benefits and Limitations

Digital repository systems are a crucial component of modern higher education, providing students with platforms for storing, organising, and accessing academic

content. These technologies include learning management systems, cloud-based storage, and digital note repositories, all of which have grown in importance as institutions transition to mixed and wholly digital learning settings. The research consistently emphasises that repository systems provide considerable benefits in terms of accessibility, flexibility, and scalability, allowing students to interact with learning resources across devices and environments (Bond et al., 2018).

One of the key advantages of repository systems is their ability to help organise and centralise information. Arden et al. (2024) notes that students are increasingly relying on digital repositories to help organise academic content, readings and personal notes. These systems allow students to efficiently store key information during lectures and support the transition from paper-based study. From an institutional perspective, repositories also provide consistency in content distribution and reduce the risk of information loss.

While repository systems are good at preserving knowledge, the literature indicates that they frequently serve as passive containers rather than active learning environments. Sillence et al. (2022) emphasises that students frequently accumulate large amounts of academic data in digital repositories without clear organisational systems. Their findings indicate that this accumulation can lead to information overload, making it harder for students to identify relevant material while revising. As repositories expand over time, their value may decrease, especially if students rely on folder structures or inconsistent naming conventions.

A further limitation identified in the literature is lack of support for meaningful engagement with stored content. Bond et al. (2018) argues that many digital repositories prioritise access and efficiency over effectiveness. Repository systems are often designed to distribute content rather than facilitate how students digest, revisit or apply the content. As a result, students may engage in surface level study habits like reviewing notes rather than deeper forms of learning that promote long term retention.

Research on study habits reinforces this concern. Arden et al. (2024) argues that while students change their note-taking practices to digital contexts, these changes do not necessarily result in better learning results. Although digital notes and repositories can indicate quick information collection, the learning value of recorded content may be restricted if organised opportunities for review and reflection are not provided. This highlights a key issue within repository-based systems which is that they excel at collecting information but often fail to support subsequent stages of learning.

The literature also implies that repository systems are mostly unrelated to evidence-based learning methodologies. According to Trumble et al. (2023) and Firth et al. (2021) strategies such as retrieval practice and spread learning are particularly successful in promoting long-term retention. However, these strategies are rarely embedded into repository platforms. Instead, students often rely on separate digital tools for revision

activities, resulting in a disconnect between where material is kept and how it is learned.

This separation can reduce the effectiveness of both systems. Trumble et al. (2023) argues that retrieval-based learning is most successful when students effortlessly connect recalled knowledge to its original academic context. When repositories and revision tools operate independently, students must manually re-establish these connections, which increases cognitive effort and reduces efficiency. Sillence et al. (2022) also states that students frequently struggle to maintain consistent organisational structures across numerous digital platforms, leading to fragmentation.

Synthesising these findings reveals that repository systems offer clear benefits in terms of access, storage and flexibility, but they also have substantial limitations in facilitating good learning habits. While repositories effectively manage academic content, they frequently fail to encourage active participation, structured revision or contextual linkage between materials. According to the study, these constraints are caused by system designs that prioritise storage over learning, rather than a lack of student motivation (Bond et al., 2018; Arden et al., 2024).

This analysis offers a compelling argument for the planned investigation. Rather of replacing repository systems, the literature encourages the creation of improved repository systems that incorporate learning-oriented features. Many of the issues raised might be addressed by an integrated repository that promotes organisation while also allowing for retrieval-based learning and contextual linking. These findings immediately inform the planned project and lay the foundation for the requirements and design considerations described in following chapters.

3.4 Flashcard benefits and limitations

Flashcards remain one of the most prominent revision tools in higher education because they correspond well with evidence-based learning principles, especially retrieval practice and separated practice. Rather than providing passive review, flashcards encourage learners to develop an answer, which strengthens memory and improves future performance. This is reflected in the health-professions education literature, where retrieval and distributed practice are frequently linked with greater academic achievements with less active study approaches. (Trumble et al., 2023)

One significant advantage of flashcards is that they provide a low-barrier method for operationalising retrieval learning. Students can convert lecture topics into brief prompts that

can be practiced repeatedly, making flashcards useful for high-volume material and time-sensitive revision. According to Lu, Farhat and Beck Dallaghan (2021) research on Anki-based learning, using a spaced-repetition flashcard system is associated with better perceived retention and assessment outcomes in medical education environments, supporting the argument that flashcards can serve as an efficient supplement to formal teaching. Furthermore, curriculum-linked electronic flashcards have been demonstrated to be viable and well-received in medical programs, with usage correlated with outcomes in some circumstances, demonstrating that students interact with flashcards when they are associated with course requirements. (Sun et al., 2021)

Another advantage of using flashcards digitally is that they allow for spaced repetition and effective scheduling. According to Carpenter, Pan and Butler (2022) research evaluations, spacing and retrieval are effective methods, but they are generally underutilised unless students employ tools that manage repetition over time. Flashcard systems that provide information at increasing intervals assist in transferring this research into habitual practice, particularly when students are handling numerous modules and require a constant review routine. In practice, this matters because students frequently fail to self-regulate revision without guidance, flashcard scheduling reduces the planning effort by determining what to study next based on past performance.

Flashcards can also be adaptable in terms of question structure, and literature indicates that format selection influences how much learning is absorbed. Prompts might need simple answers, extended explanations, or recognition-based selection; thus, retrieval practice involves more than just one exercise. Comparing different retrieval formats demonstrates that the design of retrieval questions may influence retention outcomes, demonstrating that the instructional value of flashcards is determined not just by their use but also by how they are structured. This is crucial for the project since it includes assistance for developing high-quality cards rather of presuming that any flashcard would elicit great learning. (van Wijk et al., 2024)

Despite these benefits, the literature identifies significant limitations. According to Carpenter, Pan and Butler (2022) one limitation is that flashcards may encourage fragmented and decontextualised learning by reducing complicated topics to specific information. While retrieval exercise improves memory for the studied material, flashcards may not automatically assist conceptual comprehension unless prompts are constructed to encourage explanation, application, or integration. Reviews of spacing and retrieval stress that these methods are effective, but their effectiveness is dependent on the alignment of the learning aim. In other words, flashcards are effective for recall but may be ineffective for higher-order learning unless well designed.

A second limitation is the time and skills needed for developing useful flashcards. According to Lu, Farhat and Beck Dallaghan (2021) students typically develop decks hastily, which might result in poorly written prompts, ambiguous answers, or cards that are too simple. Where studies show beneficial results, students often utilise flashcards consistently and in ways that correspond to course content, this indicates that advantages may be reduced when students generate low-quality cards or neglect review habits. This is a design implication which should allow systems both card storage and quality control.

A third limitation is an inconsistency between assessment and educational structure. Flashcards are most successful when the prompts correspond to what students are expected to know and do in exams. According to Sun et al. (2021) research on curriculum-related

flashcards, consumption and effectiveness increases when flashcards are explicitly linked to taught material and learning outcomes. Without such alignment, flashcards may become a study tool that competes with rather than reinforces course information, reducing motivation and long-term use.

Finally, although retrieval practice and spaced repetition are widely recommended, research shows that children do not always use these methods efficiently without guidance and feedback. According to Carpenter, Pan and Butler (2022) reviews, even when learners are aware of beneficial tactics, they frequently underuse them, indicating a gap between research and everyday practice that educational systems could help bridge through reminders, scheduling, and performance feedback. This emphasises the necessity for tool design that promotes consistency rather than depending solely on self-discipline.

Overall, the research supports flashcards as an effective device for retrieval practice and spaced learning, especially when digital technologies are used to help structure reviews and sustain engagement. At the same time, flashcards have constraints with decontextualisation, card quality, and alignment with learning aims. These findings support a design approach in which flashcards are integrated into a larger learning workflow, such as linking cards back to the original notes or sources to preserve context, promoting high-quality prompt design, and incorporating spaced scheduling to make evidence-based practice easier to maintain.

3.5 Conclusions

This literature review has identified multiple important insights that will directly guide the proposed project's path. The studied research consistently shows that, while students extensively utilise digital technologies to save and organise academic resources, they frequently suffer with information overload, fragmented systems, and ineffective revision procedures. Repository systems have obvious advantages in terms of accessibility and centralisation, but they usually serve as passive storage places that does not actively encourage learning or long-term retention.

In contrast, the literature on flashcard-based learning shows substantial support for tactics that actively promote memory and learning. In comparison to passive study methods, retrieval practice and spaced repetition have been proven to boost long-term retention (Trumble et al., 2023). Flashcards are thus an effective learning aid when utilised in a structured and purposeful manner. However, the research identifies significant limitations, such as the possibility of surface-level learning and insufficient contextual linkages between revision materials and original academic sources (Firth et al., 2021).

Taken together, the evidence reveals that a major issue is not a lack of good study tools, but rather a lack of synergy among them. Although repository systems and flashcard tools address complimentary parts of the learning process, they are often viewed as different solutions. This division leads to fragmented study routines and increases cognitive demands on students.

This information will be used to the project's next stages. The requirements analysis will concentrate on identifying qualities that facilitate both effective organising of academic resources and active learning via retrieval practice. The design phase will focus on preserving obvious linkages between stored materials and flashcard-based revision information, as well

as resolving contextual gaps highlighted in the literature. Finally, the prototype step will investigate how evidence-based learning techniques might be integrated into a repository environment rather than being developed as separate tools.

Overall, this literature analysis offers a clear explanation for the proposed project and serves as a solid research basis for the requirements, design, and prototyping decisions described in following chapters.

4. Product Research

4.1 Review/Comparison

To reinforce the proposed system's usability focused design, several of the existing digital platforms commonly used among university students were reviewed and compared. These platforms were chosen because they are frequently used solutions for academic resource management or flashcard-based revision and therefore reflect the tools that students currently use in practice. The comparison focuses on usability features rather than technical implementation, with the aim of identifying strengths and limitations that impact student study habit.

The platforms chosen for comparison include repository-based systems such as Google Drive and Microsoft OneNote, as well as flashcard-based learning application such as Anki and Quizlet. These platforms were selected because of their extensive usage in higher education and their representation of various techniques to storing, organising and updating academic materials.

To ensure a consistent structure, a set of usability criteria was defined, informed by themes identified in the literature review and established usability principles. The criteria include ease of navigation, organisation and structure, support for learning activities, contextual linkage, cognitive load, and user effort required for ongoing use. These criteria address typical difficulties raised in the literature, such as information overload, fragmented workflows, and inefficient revision methods (Sillence et al., 2022; Arden et al., 2024).

Repository based platforms, such as Google Drive, offer significant usability benefits in terms of accessibility and centralisation. They allow students to store large amounts of academic content and access it across several devices and platforms. However, usability concerns frequently emerge when repositories expand over time. Folder based structure can be complex and inconsistent, making it difficult for users to find certain files quickly. Without enforced structure or learning based features, these platforms are more like passive storage spaces than tools that actively support revision.

Similarly, Microsoft OneNote provides flexible note-taking and organisational tools, such as notebooks and sections, which may be used to collect and organise lecture notes. OneNote's usability features include rapid information capture and cross-device

synchronisation. However, its emphasis on free form note storage, it does not support organised editing or active recall. Users must manually revisit notes, which might lead to passive review rather active learning.

Anki and other flashcard-based systems excel at supporting retrieval practice and spaced repetition. Anki's scheduling system reduces cognitive effort by determining when cards should be reviewed, resulting in more consistent engagement over time. According to research students who use Anki can improve their retention by using flashcards on a regular and appropriate basis. However, from a usability aspect, Anki can be difficult to understand for new users, and its UI may be seen as unintuitive discouraging adoption (Lu et al., 2021; Santhosh et al., 2024).

Quizlet's interface is more user-friendly and straightforward interface, making flashcard-based revision more accessible. Its usability strengths include easy card creation and multiple revision modes. However, its default emphasises on recognition-based tasks and inadequate contextual relationship to original learning materials may restrict its usefulness for deeper learning. Furthermore, the separation of flashcards and source materials forces students to actively maintain context between platforms.

When evaluated against the established usability criteria, a distinct pattern emerges. Repository platforms excel at organisation and accessibility, whereas flashcard platforms excel at active learning and revision. However, neither category can sufficiently handle both functions inside a single, cohesive system. Students must consequently handle several tools, which increases cognitive load and reduces overall efficiency.

This comparison highlights a usability deficiency that directly influences the proposed solution. By combining repository functionality with flashcard-based learning and preserving clear links between stored resources and revision material, the proposed platform aims to integrate the usability benefits of existing solutions while addressing their limitations. These insights guide the requirements and design decisions.

5. Legal, Social, Ethical and Professional issues and Considerations

Legal:

This project must comply with all legal laws like data protection and intellectual property rules and regulations, particularly when dealing with user-generated academic work. According to Liu and Khalil (2023), compliance with the General Data Protection Regulation is critical for systems that gather or handle academic data since it provides openness and permission. According to Welford (2025), GDPR establishes concepts such as lawful processing, data minimisation, and user control

over personal information, which is critical for ensuring trust and compliance in digital academic systems. Similarly, the UK Government (2018) specifies how the Data Protection Act 2018 must maintain personal data and is protected under UK law, supplementing the GDPR's

framework for responsible data management. As a result, the repository will incorporate privacy warnings and only collect data required for academic reasons. Copyright compliance will also be maintained by allowing users to contribute properly referenced academic writings that must be authorised by an administrator, preventing legal violation of published information.

Social:

From a social aspect, the system encourages students to collaborate and share their experiences by also requesting permission for specific sources to be added. As highlighted by Khalil et al. (2023), educational technologies can enhance peer interaction and motivation when developed with inclusiveness with accessibility in mind. This system supports social equality by being web-based and accessible across numerous platforms, eliminating digital barriers for students from diverse backgrounds, and including accessibility features like as screen reader compatibility, configurable text size, and clear visual layouts. These design considerations enable students with visual or reading challenges to interact with academic information more freely and efficiently, ensuring that the platform is accessible to all users.

Ethical:

This project prioritises academic integrity, fairness and ethical technology usage in an educational environment. According to Huang et al. (2025), ensuring transparency, privacy and accountability in digital education platforms is critical for preventing unethical activity and increasing user confidence. In compliance with these findings, the proposed repository would guarantee that all user data is anonymised and handled appropriately, while also including plagiarism prevention measures and citation prompts to promote academic integrity. Furthermore, the system's design will ensure algorithmic fairness by preventing bias in flashcard production or source evaluation while promoting equitable learning chances for all users.

Professional:

From a professional viewpoint, this project adheres to recognised computing standards for integrity, accountability, and ethical responsibility. Insights from O'Shea (2024), emphasise the importance of incorporating ethics and professional conduct into computing curricula, reinforcing the need for projects like this to adhere to ethical best practices from the start. Additionally, Blaise et al. (2024) highlights that well-structured ethical frameworks assist information systems professionals in navigating complicated problems such as privacy, data protection, and responsible system design. By incorporating these principles into the building of the system, including secure processing of

academic data, fair depiction of material, and accessible design, the project displays commitment to professional standards and assures responsibility.

6. Requirements

6.1 Analysis of requirements

The project's requirements analysis is determined by the project's goal, set goals, literature study findings, and insights acquired by examining existing repository and flashcard-based educational tools. The major goal of the project is to develop a web-based tool that combines a research repository with flashcard capabilities to help students organise academic materials, practise active recall, and improve knowledge retention while decreasing cognitive overload. As a result, the system requirements must promote both effective information management and evidence-based learning techniques.

The literature review identified many important issue areas that have a direct impact on the requirements analysis. Studies on student usage of digital tools often highlight difficulties such as information overload, fragmented study processes, and ineffective revising practices. Repository systems were found to be useful for storing and accessing academic content, but they frequently lacked aspects that actively promote learning and retention. Flashcard-based systems, on the other hand, allowed for retrieval practice and spaced learning but were often separated from source information, restricting contextual knowledge. These findings suggest that the system should satisfy both organisational and learning-related criteria in an integrated way.

A review of existing repository and flashcard platforms supplemented the requirements analysis. Popular repository technologies prioritise flexibility and storage capacity but often rely largely on manual organising via folders or tags, which can be difficult to manage over time. Flashcard platforms, while useful for revision, sometimes require users to navigate between systems and manually handle linkages between revision material and original sources. This division raises cognitive burden and creates inefficiencies in the study process. As a result, a key requirement highlighted is the seamless integration of stored academic materials with flashcard-based revision content.

In addition, user-related concerns are important in requirements analysis. The system is designed for use by students with many different levels of technical knowledge. Therefore, usability and simplicity of navigation are critical. The requirements must thus provide straightforward procedures for adding resources, structuring content, creating flashcards, and evaluating material. Complex structure or excessive manual input might discourage consistent use, reducing the system's learning advantages. This is consistent with the project's aim of decreasing information overload rather than contributing to it.

Security and data management issues were also examined as part of the requirements process. As the system contains user accounts and academic data storage, it is necessary to have secure user authentication, regulated data access, and suitable administration of system roles such as administrators and everyday users. These factors guarantee that the system is appropriate for real-world usage and meets the required criteria for web-based applications.

The project's implementation constraints and development style also influence requirement analysis. The system is a web-based application built using PHP and HTML, with a relational database as backend. This effects needs for data structure, speed, and scalability. The

database, for example, must provide efficient interactions between people, resources, and flashcards in order to enable quick retrieval and contextual links. Similarly, the system architecture must accommodate future expansion, allowing new features to be introduced without requiring extensive redesign.

Finally, the requirements analysis is iterative and based on expected user input. This process's feedback is meant to modify and prioritise requirements, ensuring that the system meets actual user demands rather than theoretical assumptions. This user-centred approach guarantees that the criteria are both academically solid and practically applicable.

To summarise, the analysis of requirements uses literature discoveries, product research, user expectations, and technological restrictions to establish what the system must do. This analysis serves as the basis for the functional and non-functional requirements described in the next sections, ensuring that future design and implementation decisions are consistent with the project's overall aim and objectives.

6.2 Requirements elicitation

6.3 Functional Requirements

6.4 Non-Functional; Requirements

7. Design

8. Prototype

8.1 Review of technology

9. Bibliography

10. Appendices

11. References

Arden, B., Norris, J., Cole, S. and Gamble, S.C. (2024). Digital notetaking in lectures: how students adapt to a multi-faceted university learning environment. *Cogent education*, 11(1). doi:<https://doi.org/10.1080/2331186x.2024.2373552>

Blaise, O.O., Aaron, I., Alfred, U. and Afolarin Amusa (2024). Evaluating the Ethical Frameworks of Information Security Professionals: A Comparative Analysis. *Asian Journal of Computer Science and Technology*, 13(2), pp.61–66.
doi:<https://doi.org/10.70112/ajcst-2024.13.2.4289>

Bond, M., Marín, V.I., Dolch, C., Bedenlier, S. and Zawacki-Richter, O. (2018). Digital transformation in German higher education: student and teacher perceptions and usage of digital media. *International Journal of Educational Technology in Higher Education*, 15(1). doi:<https://doi.org/10.1186/s41239-018-0130-1>.

Carpenter, S.K., Pan, S.C. and Butler, A.C. (2022). The science of effective learning with spacing and retrieval practice. *Nature Reviews Psychology*, [online] 1(9), pp.496–511. doi:<https://doi.org/10.1038/s44159-022-00089-1>.

Firth, J., Rivers, I. and Boyle, J. (2021). A Systematic Review of Interleaving as a Concept Learning Strategy. *Review of Education*, [online] 9(2). doi:<https://doi.org/10.1002/rev3.3266>.

Friess, E. (2022). Scrum in Classroom Collaborations: A Quasi-Experimental Study. *Journal of Business and Technical Communication*, p.105065192211218. doi:<https://doi.org/10.1177/10506519221121817>

Huang, C.L., Shao, X., Wu, C. and Yang, S.C. (2025). Navigating the digital learning landscape: insights into ethical dilemmas and academic misconduct among university students. *International Journal of Educational Technology in Higher Education*, 22(1). doi:<https://doi.org/10.1186/s41239-025-00516-2>

Khalil, M., Slade, S. and Prinsloo, P. (2023). Learning analytics in support of inclusiveness and disabled students: a systematic review. *Journal of Computing in Higher Education*, [online] 36. doi:<https://doi.org/10.1007/s12528-023-09363-4>

Liu, Q. and Khalil, M. (2023). Understanding privacy and data protection issues in learning analytics using a systematic review. *British Journal of Educational Technology*, 54(6), pp.1715–1747. doi:<https://doi.org/10.1111/bjet.13388>

Lu, M., Farhat, J.H. and Beck Dallaghan, G.L. (2021). Enhanced Learning and Retention of Medical Knowledge Using the Mobile Flash card Application Anki. *Medical Science Educator*, [online] 31(6), pp.1975–1981. doi:<https://doi.org/10.1007/s40670-021-01386-9>

O'Shea, K. (2024). Evaluating UK academics' perspectives of ethics education within computer science degree programmes: a preliminary insight. *International Journal of Ethics Education*. doi:<https://doi.org/10.1007/s40889-024-00205-0>

Santos, M.F., Filipe, R.A. and Cunha, J.C. (2024). From Traditional to Agile Methodologies in Software Project Management Education: A Case Study. *From Traditional to Agile Methodologies in Software Project Management Education*, pp.491–497. doi:<https://doi.org/10.1145/3702163.3702460>

Sillence, E., Dawson, J.A., McKellar, K. and Neave, N. (2022). How do students use digital technology to manage their university-based data: strategies, accumulation difficulties and feelings of overload? *Behaviour & Information Technology*, 42(14), pp.1– 10. doi:<https://doi.org/10.1080/0144929x.2022.2126948>

Sun, M., Tsai, S., Engle, D.L. and Holmer, S. (2021). Spaced Repetition Flashcards for Teaching Medical Students Psychiatry. *Medical Science Educator*, 31(3). doi:<https://doi.org/10.1007/s40670-021-01286-y>.

Trumble, E., Lodge, J.M., Mandrusiak, A. and Forbes, R. (2023). Systematic review of distributed practice and retrieval practice in health professions education. *Advances in Health Sciences Education*, 29, pp.689–714. doi:<https://doi.org/10.1007/s10459-023-10274-3>.

UK Government (2018). *Data Protection Act*. [online] Gov.uk. Available at: <https://www.gov.uk/data-protection> [Accessed 15 Oct. 2025].

van Wijk, E.V., de Jonge, M., van Blankenstein, F.M., Janse, R.J. and Langers, A.M.J. (2024). The battle of question formats: a comparative study of retrieval practice using very short answer questions and multiple choice questions. *BMC Medical Education*, 24(1). doi:<https://doi.org/10.1186/s12909-024-06538-0>.

Varkey Nadakkavukaran Santhosh, Coutinho, D., Ankola, A.V., Yuvarani Kandasamy Parimala, Siva Shankkari and Kavitha Ragu (2024). Effectiveness of spaced repetition learning using a mobile flashcard application among dental students: A randomized controlled trial. *Journal of dental education*. doi:<https://doi.org/10.1002/jdd.13561>

Wolford, B. (2025). *What is GDPR, the eu's new data protection law?* [online] GDPR.EU. Available at: <https://gdpr.eu/what-is-gdpr/> [Accessed 15 Oct. 2025].